§0. Backgrounds

· Basic Knowledge about Sets

(X, Y: sets)

- **Equality between sets**
  : $X = Y \overset{\text{def}}{\iff} X \subseteq Y \ \wedge \ Y \subseteq X$.
- **Intersection**: $X \cap Y \equiv \{x \mid x \in X \ \wedge \ x \in Y\}$,
  Union: $X \cup Y \equiv \{x \mid x \in X \ \vee \ x \in Y\}$.
- X and Y are disjoint $\overset{\text{def}}{\iff} X \cap Y = \varnothing$.
- **Subtraction**: $X \setminus Y \equiv \{x \mid x \in X \ \wedge \ x \notin Y\}$.
- **Distribute Law**
  · $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.
  · $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$.
- **De Morgan's Law**
  · $X \setminus (Y \cap Z) = (X \setminus Y) \cup (X \setminus Z)$.
  · $X \setminus (Y \cup Z) = (X \setminus Y) \cap (X \setminus Z)$.
- **Equality between Ordered pairs**
  : $(x, y) = (x', y') \overset{\text{def}}{\iff} x = x' \ \wedge \ y = y'$
  ($x, y, x', y' \in X$)
- **Cartesian Product**
  : $X \times Y \equiv \{(x, y) \mid x \in X \ \wedge \ y \in Y\}$
  ($\varnothing \notin \{X, Y\}$)
- **Power set**
  · Power set of X: $\mathscr{P}(X) \equiv \{U \mid U \subseteq X\}$.
  · Intersection of $\mathscr{U} \subseteq \mathscr{P}(X)$
  : $\bigcap \mathscr{U} \equiv \bigcap_{U \in \mathscr{U}} U \equiv \{x \mid \forall U \in \mathscr{U}, \ x \in U\}$.
  · Union of $\mathscr{U} \subseteq \mathscr{P}(X)$
  : $\bigcup \mathscr{U} \equiv \bigcup_{U \in \mathscr{U}} U \equiv \{x \mid \exists U \in \mathscr{U}, \ x \in U\}$.

· Basic Knowledge about Functions

(f: function, X, Y: nonempty sets)

- **Definition of Function**
  : For a set $F \subseteq X \times Y$,
  if $\forall x \in X$, $\exists y \in Y$ s.t. $(x, y) \in F$,
  then $f := (X, Y, F)$ is a function from X into Y.
- **Expression of Function**
  : For a function $f = (X, Y, F)$ s.t. $(x, y) \in F$, we write
  $f\colon \begin{matrix} X \longrightarrow Y, \\ x \longmapsto y, \end{matrix}$ or $y = f(x)$.
  Also, for $U \subseteq X$, $f(U)$ is the set $\{f(x) \mid x \in U\}$.
- **Sets Around Function**
  : For a function $f\colon X \longrightarrow Y, \ x \longmapsto y$,
  · X is the domain of f.
  · Y is the codomain of f.
  · $f(X)$ $(= \{f(x) \mid x \in X\})$ is the range of f.
  · $G(f) := \{(x, f(x)) \mid x \in X\}$ is the graph of f.
- **Operation between Functions (1)**
  ($f\colon X \longrightarrow \mathbb{R}$, $g\colon Y \longrightarrow \mathbb{R}$, $x \in (X \cap Y)$)
  · Addition/Subtraction
  : $f \pm g\colon X \cap Y \longrightarrow \mathbb{R}$ s.t.
  $(f \pm g)(x) \equiv f(x) \pm g(x)$.
  · Multiplication
  : $fg\colon X \cap Y \longrightarrow \mathbb{R}$ s.t.
  $(fg)(x) \equiv f(x) \cdot g(x)$.
  · Division
  : $\dfrac{f}{g}\colon \{x \in X \cap Y \mid g(x) \neq 0\} \longrightarrow \mathbb{R}$ s.t.
  $(f/g)(x) \equiv \dfrac{f(x)}{g(x)}$.
- **Types of Functions**
  ($f\colon X \longrightarrow Y$, $x_1, x_2 \in X$)
  · f is injective if $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
  · f is surjective if $f(X) = Y$.
  · f is bijective if f is injective and also surjective.
- **Operation between Functions (2)**
  ($f\colon X \longrightarrow Y$, $g\colon Z \longrightarrow W$, $(x, y) \in G(f)$)
  · Inverse Function
  : $f^{-1}\colon Y \longrightarrow X$ s.t. $f^{-1}(y) = x$ exists
  if and only if f is bijective.
  · Composed Function
  : $g \circ f\colon X \longrightarrow W$ s.t. $(g \circ f)(x) = g(f(x))$ exists
  if $f(X) \subseteq Z$.

§1.1. Metric Fcn. & Open Ball

· Metric Functions

- Definition

: For a nonempty set X and $\mathbb{R}^+ = [0, +\infty)$, a function

$$d: X \times X \longrightarrow \mathbb{R}$$

is a metric on X if it satisfies following 3 properties:

(M1) $d(x, y) = 0$ iff $x = y$,

(M2) $d(x, y) = d(y, x)$,

(M3) $d(x, z) \leqslant d(x, y) + d(y, z)$

for all $x, y, z \in X$.

- A metric space is a pair (X, d) or $\underline{X}$.
- We call $x \in X$ the point of X.
- The distance of x, y is the image of d, i.e., $d(x, y)$.

- Theorems

$[(X, d)$: m.space, $x, y, z, w \in X$, $n \in \mathbb{N}$, $1 \leqslant i \leqslant n+1]$

· $d(x_i, x_{n+1}) \leqslant \sum_{j=i}^{n} d(x_j, x_{j+1})$.

· $|d(x, z) - d(y, z)| \leqslant d(x, y)$

· $|d(x, y) - d(z, w)| \leqslant d(x, z) + d(y, w)$

· Examples of Metric Functions

- d_E: Euclidean metric on $\mathbb{R}^n$ $(n \in \mathbb{N})$

: For $x := (x_1, x_2, \ldots, x_n)$ and $y := (y_1, y_2, \ldots, y_n)$,

$$d_E: \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R} \quad \text{s.t.}$$

$$d_E(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \quad \text{for all } x, y \in \mathbb{R}$$

is an Euclidean metric on $\mathbb{R}^n$.

- d_D: Discrete metric: 이산 거리 함수

: For a nonempty set X,

$$d_D: X \times X \longrightarrow \mathbb{R} \quad \text{s.t.}$$

$$d_D(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y \end{cases} \quad \text{for all } x, y \in X$$

is a discrete metric on X.

- Metric on V

: For a vector space V and the norm operator $\|\cdot\|$,

$$d: V \times V \longrightarrow \mathbb{R} \quad \text{s.t.}$$

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| \quad \text{for all } \mathbf{u}, \mathbf{v} \in V$$

is a metric on V.

$$\left(\|\cdot\|: \begin{array}{l} (\text{i}) \ \|\mathbf{v}\| = 0 \Longleftrightarrow \mathbf{v} = \mathbf{0} \\ (\text{ii}) \ \|\mathbf{u} + \mathbf{v}\| \leqslant \|\mathbf{u}\| + \|\mathbf{v}\| \\ (\text{iii}) \ \|k\mathbf{v}\| = |k| \cdot \|\mathbf{v}\| \end{array} \right)$$

- Supremum Metric on $B(X)$

: For $X \neq \varnothing$ and $B(X) \equiv \{f \mid f: X \longrightarrow \mathbb{R},\ f: \text{bdd.}\}$,

$$d: B(X) \times B(X) \longrightarrow \mathbb{R}^+ \quad \text{s.t.}$$

$$d(f, g) = \sup_{x \in X} |f(x) - g(x)| \quad \text{for all } f, g \in B(X)$$

is a supremum metric on $B(X)$.

- d_M: Maximum Metric on $\mathbb{R}^n$

: For $x := (x_1, x_2, \ldots, x_n)$ and $y := (y_1, y_2, \ldots, y_n)$,

$$d_M: \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}^+ \quad \text{s.t.}$$

$$d_M(x, y) = \max\{|x_i - y_i| : 1 \leqslant i \leqslant n\}$$

is a maximum metric on $\mathbb{R}^n$.

- d_S: Sum Metric on $\mathbb{R}^n$

: For $x := (x_1, x_2, \ldots, x_n)$ and $y := (y_1, y_2, \ldots, y_n)$,

$$d_S: \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}^+ \quad \text{s.t.}$$

$$d_S(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

is a sum metric on $\mathbb{R}^n$.

- Inequality Between d_E, d_M, d_S

: For $x, y \in \mathbb{R}^n$, $d_M(x, y) \leqslant d_E(x, y) \leqslant d_S(x, y)$.

· Open Ball

- Definition

: A set $S^d(x, \epsilon) \equiv \{y \in X : d(x, y) < \epsilon\}$ is an open ball with the center x and radius ϵ relative to d, for a m.space (X, d), given $\epsilon > 0$, and a point $x \in X$.

- Examples

· $S^{d_E}(0, 1) = (-1, 1)$ in $(\mathbb{R}, d_E)$;

· $S^{d_E}((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} < 1\}$;

· $S^{d_S}((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 : |x| + |y| < 1\}$;

· $S^{d_M}((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 : |x| < 1, |y| < 1\}$.

§1.2. Important Points & Subset of Metric Spaces

· Inner Points/Sets

- Inner Points

: $x \in X$ is an inner point of a set $B \subseteq X$, if $\exists \epsilon > 0$ s.t. $S(x, \epsilon) \subseteq B$.

- Inner Sets

: For $B \in X$, $\mathring{B}$ is an inner set of A, where $\mathring{B} \equiv \{x \in X : x \text{ is an inner point of } B\}$.

Thus, $x \in \mathring{B}$ implies that x is an inner point of B.

· Open Set

- Open Sets

: For $B \subseteq X$, B is an open set, if $B = \mathring{B}$.

If B: open set, then $\forall x \in B$, $\exists \epsilon > 0$ s.t. $S(x, \epsilon) \subseteq B$.

- Open Balls are Open Sets

: $S(x, \epsilon)$ is an open set, for all $x \in X$ and given $\epsilon > 0$.

pf.

Let $\delta = \epsilon - d(x, y)$ for $y \in S(x, \epsilon)$, and $z \in S(y, \delta)$.
Then, $d(y, z) < \delta$, hence

$$\begin{aligned} d(x, z) &\leqslant d(x, y) + d(y, z) \\ &< d(x, y) + \delta \\ &< d(x, y) + \epsilon - d(x, y) = \epsilon, \end{aligned}$$

thus, $z \in S(x, \epsilon)$, which implies $S(y, \delta) \subseteq S(x, \epsilon)$.
Therefore, we have $\exists \forall y \in S(x, \epsilon)$; $\exists \delta$ s.t. $S(y, \delta) \subseteq S(x, \epsilon)$, which explains $S(x, \epsilon)$ is an open set. ■

- Theorems (1) $(A, B \subseteq X)$

(a) $\mathring{\varnothing} = \varnothing$, $\mathring{X} = X$.

(b) $\mathring{A} \subseteq A$.

(c) $B \subseteq A$ implies $\mathring{B} \subseteq \mathring{A}$.

(d) $\widehat{A \cap B} = \mathring{A} \cap \mathring{B}$.

(e) $\mathring{\mathring{A}} = \mathring{A}$.

- Theorems (2)

(a) $\varnothing$, X are open sets.

(b) If A, B are open sets, then $A \cap B$ is so.

(c) If $\mathscr{B}$ is a set of open sets of X, $\bigcup \mathscr{B} \equiv \{x \in B : \exists B \in \mathscr{B}\}$ is an open set.

pf.

(1d) Since $A \cap B \subseteq A$ & B, we can conclude

$$\widehat{A \cap B} \subseteq \mathring{A} \quad \text{and} \quad \widehat{A \cap B} \subseteq \mathring{B}$$

by (1c), which implies $\widehat{A \cap B} \subseteq \mathring{A} \cap \mathring{B}$.

Let $x \in \mathring{A} \cap \mathring{B}$. Then, $\exists \epsilon_1, \epsilon_2 > 0$ s.t.

$$S(x, \epsilon_1) \subseteq A \quad \wedge \quad S(x, \epsilon_2) \subseteq B.$$

Put $\epsilon := \min\{\epsilon_1, \epsilon_2\}$, and let $y \in S(x, \epsilon)$. Then $y \in S(x, \epsilon_1)$ and also $y \in S(x, \epsilon_2)$, thus $y \in S(x, \epsilon_1) \cap S(x, \epsilon_2)$, which is subset of $A \cap B$.
Hence, $y \in A \cap B$, then $S(x, \epsilon) \subseteq A \cap B$, thus $\mathring{A} \cap \mathring{B} \subseteq \widehat{A \cap B}$.

Therefore, we have $\widehat{A \cap B} = \mathring{A} \cap \mathring{B}$. ■

. . . . . .

(1e) Let $x \in \mathring{A}$. Then we have

$$\begin{aligned} &\exists \epsilon > 0 \text{ s.t. } S(x, \epsilon) \subseteq A; \\ \Longrightarrow\ &\widehat{S(x, \epsilon)} \subseteq \mathring{A}; \\ \Longrightarrow\ &S(x, \epsilon) \subseteq \mathring{A}; \\ \Longrightarrow\ &\mathring{A} \text{ is a open set}; \\ \Longrightarrow\ &\mathring{\mathring{A}} = \mathring{A}. \end{aligned}$$ ■

. . . . . .

(2c) By (1b), we have $\widehat{\bigcup \mathscr{B}} \subseteq \bigcup \mathscr{B}$. ····· (1)

Let $x \in \bigcup \mathscr{B}$. Then $x \in B$ for a open set $B \in \mathscr{B}$, thus $S(x, \epsilon) \subseteq \mathscr{B}$ for some $\epsilon > 0$.
Then, $\exists \epsilon > 0$ s.t. $S(x, \epsilon) \subset B \subset \bigcup \mathscr{B}$. ····· (2)

By (1) and (2), we have $\widehat{\bigcup \mathscr{B}} = \bigcup \mathscr{B}$, which implies $\bigcup \mathscr{B}$ is an open set. ■

- Theorem (3)

: For $B \subseteq X$, $\mathring{B}$ is the largest open set contained by B.

pf.

Let A be a open set s.t. $A \subseteq B$. Then
(i) $\mathring{B} \subseteq B$ by (1b);
(ii) $\mathring{B}$ is an open set by (1e);
(iii) $A \subseteq B \Longrightarrow \mathring{A} \subseteq \mathring{B} \Longrightarrow A \subseteq \mathring{B}$;
thus we can conclude that B is the largest open set contained in B. ■

· Distance of subsets

- Definition

: For $A,\ B \subseteq X$ where $\underline{X}$ is a metric space,
distance of A, B is defined to a function

$$\mathrm{dist}\colon \mathscr{P}(X) \times \mathscr{P}(X) \longrightarrow \mathbb{R}^+ \cup \{\infty\},$$

such that

$$\mathrm{dist}(A,\ B) \equiv \begin{cases} \infty & \text{if } A \text{ or } B = \varnothing, \\ \inf\{d(a,\ b) : a \in A,\ b \in B\} & \text{if } A,\ B \neq \varnothing. \end{cases}$$

- Properties ($\underline{X}$: a metric space, $A,\ B,\ C \subseteq X$)

(a) $\mathrm{dist}(A,\ B) = \mathrm{dist}(B,\ A)$.

(b) $A \subseteq C$ implies $\mathrm{dist}(A,\ C) \geqslant \mathrm{dist}(B,\ C)$.

(c) $A \cap B \neq \varnothing$ implies $\mathrm{dist}(A,\ B) = 0$.

pf.

(a) Let $p\colon\ A, B \neq \varnothing$, $q\colon\ a \in A,\ b \in B$. Then

$$\begin{aligned} \mathrm{dist}(A,\ B) &= \begin{cases} \infty & \text{if } \neg p, \\ \inf\{d(a,\ b) : q\} & \text{if } p \end{cases} \\ &= \begin{cases} \infty & \text{if } \neg p, \\ \inf\{d(b,\ a) : q\} & \text{if } p \end{cases} \\ &= \mathrm{dist}(B,\ A). \end{aligned}$$ ■

(b) If $B = \varnothing$; $\mathrm{dist}(C,\ B) = \mathrm{dist}(A,\ B) = \infty$.
If $B \neq \varnothing$ but $A = \varnothing$; $\mathrm{dist}(A,\ C) = \infty \geqslant \mathrm{dist}(B,\ C)$.
If $A,\ B \neq \varnothing$, then

$$\begin{aligned} \mathrm{dist}(C,\ B) &= \inf[D_C := \{d(c,\ b) : c \in C,\ b \in B\}] \\ &\leqslant \inf[D_A := \{d(a,\ b) : a \in A,\ b \in B\}] \\ &= \mathrm{dist}(A,\ B), \end{aligned}$$

since $D_C \subseteq D_A$. ■

(c) By the assumption, there exists $c \in A \cap B$, thus

$$\begin{aligned} \mathrm{dist}(A,\ B) &= \inf\{d(a,\ b) : a \in A, b \in B\} \\ &\leqslant d(c,\ c) = 0, \end{aligned}$$

which shows $\mathrm{dist}(A,\ B) = 0$. ■

· Neighbor Relation *(in short: nhb. rel.)*

- Definition

: For $A,\ B \subseteq X$, A and B are in neighbor relation,
if $\mathrm{dist}(A,\ B) = 0$.

A & B are in nhb. rel., if and only if
$\forall \epsilon > 0;\ \exists a \in A,\ b \in B$ s.t. $d(a,\ b) < \epsilon$.

- Properties ($A,\ B,\ C \subseteq X$)

(a) If A & B: nhb. rel.; $A,\ B \neq \varnothing$.

(b) If A & B: nhb. rel.; B & A: nhb. rel.

(c) If A & B: nhb. rel. $\wedge$ $B \subseteq C$;
A & C: nhb. rel.

(d) If A & $B \cup C$: nbh. rel; A & $B \vee A$ & C: nhb. rel.

(e) If $A \cap B \neq \varnothing$; A & B: nhb. rel.

pf.

(d) By the assumption, $\mathrm{dist}(A,\ B \cup C) = 0$
$\Longrightarrow \forall \epsilon > 0\ \exists x \in A,\ y \in (B \cup C)$ s.t. $d(x,\ y) < \epsilon$.
Since $y \in B$ ... (1) or $y \in C$ ... (2),

(1) $\mathrm{dist}(A,\ B) \leqslant d(x,\ y) < \epsilon \Longrightarrow \mathrm{dist}(A,\ B) = 0$,

(2) $\mathrm{dist}(A,\ C) \leqslant d(x,\ y) < \epsilon \Longrightarrow \mathrm{dist}(A,\ C) = 0$,

Thus A & B: in nhb. rel. or A & C: in nhb. rel. ■

· Closedness

- Adherent point *(in short: adh. pt.)*

· Definition

: For $A \subseteq X$ and $x \in X$, x is an adherent point of A,
if $\mathrm{dist}(x,\ A) = 0$.

· Properties ($A,\ B \subseteq X,\ x \in X$)

(a) x: adh. pt. of A iff $\forall \epsilon > 0;\ S(x,\ \epsilon) \cap A \neq \varnothing$.

(b) If x: adh. pt. of A; $A \neq \varnothing$.

(c) If x: adh. pt. of $A \wedge A \subseteq B$;
x: adh. pt. of B.

(d) If x: adh. pt. of $A \cup B$;
x: adh. pt. of $A \vee$ adh. pt. of B.

(e) If $x \in A$; x: adh. pt. of A.

(f) x: adh. pt. of A iff $\{x\}$ & A: nhb. rel.

(These are corollaries of the properties of nhb. rel.)

- Closure

· Definition

: For $A \subseteq X$, $\overline{A}$ is the closure of A,
where $\overline{A} \equiv \{x \in X : x \text{ is an adh. pt. of } A\}$.
Thus, $x \in \overline{A}$ implies that x is an adh. pt. of A.

· Properties ($A,\ B \subseteq X$)

(a) $\overline{\varnothing} = \varnothing$, $\overline{X} = X$.

(b) $A \subseteq \overline{A}$. (cf. $\mathring{A} \subseteq A$)

(c) If $A \subseteq B$; $\overline{A} \subseteq \overline{B}$.

(d) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

(e) $\overline{\overline{A}} = \overline{A}$.

pf.

(b) Let $x \in A$. Then $\mathrm{dist}(x,\, A) = 0$, thus $x \in \overline{A}$, so we have $A \subseteq \overline{A}$. ■

(d) Since $A \cup B \subseteq \overline{A} \cup \overline{B}$, $\overline{A \cup B} \subseteq \overline{\overline{A} \cup \overline{B}} = \overline{A} \cup \overline{B}$.
Also $A,\, B \subseteq \overline{A \cup B}$, thus $\overline{A},\, \overline{B} \subseteq \overline{\overline{A \cup B}} = \overline{A \cup B}$.
Hence $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$, we have $\overline{A} \cup \overline{B} = \overline{A \cup B}$. ■

(e) By (b), $\overline{A} \subseteq \overline{\overline{A}}$ is obvious. ····· (1)
Let $x \in \overline{\overline{A}}$. Then we have $\mathrm{dist}(x,\, \overline{A}) = 0$.
Then, for $\epsilon > 0$, $\exists y \in \overline{A}$ s.t. $d(x,\, y) < \epsilon/2$,
where y is an adh. pt. of A.
Thus $\mathrm{dist}(y,\, A) = 0$, $\therefore\ \exists a \in A$ s.t. $d(a,\, y) < \epsilon/2$.

$$\begin{aligned} \Longrightarrow d(x,\, a) &\leqslant d(x,\, y) + d(y,\, a) \\ &< \epsilon/2 + \epsilon/2 = \epsilon, \end{aligned}$$

which implies $\mathrm{dist}(x,\, A) = 0$. $\therefore\ x \in \overline{A}$. ····· (2)
By (1) & (2), we have $\overline{A} = \overline{\overline{A}}$. ■

- **Closed Set**
  - **Definition**
    : For $A \subseteq X$, A is closed / a closed set, if $\overline{A} = A$.

  - **Properties** ($A,\, B,\, C \subseteq X$)
    (a) $\overline{A}$ is the smallest set which A is a subset of.
    (b) $\varnothing,\, X$ are closed (and also open).
    (c) If $A,\, B$ are closed; $A \cup B$ is so. (cf. open: $\cap$)
    (d) If $\mathscr{C}$ is a set of closed sets of X;
    $\bigcap \mathscr{C} \equiv \{x \in C : \forall C \in \mathscr{C}\}$ is also closed.
    (cf. open: $\bigcup \mathscr{B} \equiv \{x \in B : \exists B \in \mathscr{B}\}$)

pf.

(a) Since
(i) $A \subseteq \overline{A}$; (ii) $\overline{\overline{A}} = \overline{A}$ are obvious;
(ii) For a closed set B such that $A \subseteq B$,
$\overline{A} \subseteq \overline{B} = B$;
$\overline{A}$ is the smallest closed set containing A. ■

(d) $\bigcup \mathscr{C} \subseteq \overline{\bigcup \mathscr{C}}$ is obvious. (1)

$$\begin{aligned} &\bigcup \mathscr{C} \subseteq C \quad \text{for all } C \in \mathscr{C}, \\ \Longrightarrow\ &\overline{\bigcup \mathscr{C}} \subseteq \overline{C} \quad \text{for all } C \in \mathscr{C}, \\ \Longrightarrow\ &\overline{\bigcup \mathscr{C}} \subseteq \bigcup_{C \in \mathscr{C}} C = \bigcup \mathscr{C}. \quad \cdots\cdots (2) \end{aligned}$$

By (1) & (2), we conclude $\overline{\bigcup \mathscr{C}} = \bigcup \mathscr{C}$. ■

  - **Relation of Inner & Closure**
    - For $B \subseteq X$ and $x \in X$,
      $\Longrightarrow\ \mathring{B} = X \setminus \overline{X \setminus B}$.
      (x: inner pt. of B iff $x \notin \overline{X \setminus B}$.)

pf.

$$\begin{aligned} x \in \mathring{B} &\Longleftrightarrow \exists \epsilon > 0 \text{ s.t. } S(x,\, \epsilon) \subseteq B; \\ &\Longleftrightarrow S(x,\, \epsilon) \cap (X \setminus B) = \varnothing; \\ &\Longleftrightarrow x \notin X \setminus B. \end{aligned}$$ ■

    - $B = \mathring{B}$ iff $X \setminus B = \overline{X \setminus B}$.
      (B is open iff $X \setminus B$ is closed.)

pf.

$$\begin{aligned} B = \mathring{B} &\Longleftrightarrow X \setminus B = X \setminus \mathring{B}; \\ &\Longleftrightarrow X \setminus B = X \setminus (X \setminus \overline{X \setminus B}); \\ &\Longleftrightarrow X \setminus B = \overline{X \setminus B}. \end{aligned}$$ ■

  - **Theorems** ($A,\, B \subseteq X$)
    - If $\overline{A} \cap \overline{B} \neq \varnothing$, then $A,\, B$ are in nhb. rel.
      ($\Longleftrightarrow \overline{A} \cap \overline{B} \neq \varnothing$ implies $\mathrm{dist}(A,\, B) = 0$.)
    - $\mathrm{dist}(A,\, B) = \mathrm{dist}(\overline{A},\, B)$.
      ($\Longrightarrow \mathrm{dist}(A,\, B) = \mathrm{dist}(\overline{A},\, \overline{B})$.)

- **Neighborhood** *(in short: nbd.)*
  - **Definitions** ($A,\, U \subseteq X$)
    - **Neighborhood of a point**
      : U is a neighborhood of x, if $x \in \mathring{U}$,
      i.e., $\exists \epsilon > 0$ s.t. $S(x,\, \epsilon) \subseteq U$.
    - **Neighborhood of a set**
      : U is a neighborhood of A, if $a \in \mathring{U}\ \ \forall a \in A$,
      i.e., for each $a \in A$; $\exists \epsilon > 0$ s.t. $S(a,\, \epsilon) \subseteq U$.
    - **Uniform Neighborhood**
      : U is a uniform neighborhood of A,
      if $\exists \epsilon > 0$ s.t. $S(a,\, \epsilon) \subseteq U\ \ \forall a \in A$.

  (nbd. vs. uniform nbd.
  : normal nbd. requires many ϵ for each $a \in A$,
  but unifrom nbd. requires only single ϵ for all $a \in A$.)

- **Theorems** (A, $U \subseteq X$, $x \in X$)
  - U is nbd. of $A \implies A \subseteq U$.
  - $a \in \mathring{A} \iff A$ is nbd. of a.
  - A is open $\iff A$: nbd. of $\forall a \in A$.

pf.

($\Longrightarrow$) Assume A is open, and let $a \in A$.
then $a \in \mathring{A}$, thus A is nbd. of A. □

($\Longleftarrow$) Let $a \in A$, and assume A is nbd. of a.
Then, a is an inner pt. of A, i.e., $a \in \mathring{A}$,
which implies $A \subseteq \mathring{A}$.
And $\mathring{A} \subseteq A$ is obvious, thus $\mathring{A} = A$,
i.e., A is open. ■

- U is nbd. of $x \iff x \notin \overline{X \setminus U}$.

pf.

($\Longrightarrow$) Assume U is nbd. of x.
$\implies \exists \epsilon > 0$ s.t. $S(x, \epsilon) \subseteq U$.
$\implies S(x, \epsilon) \cap (X \setminus U) = \varnothing$.
$\implies x \notin \overline{X \setminus U}$. □

($\Longleftarrow$) Conversely, assume $x \notin \overline{X \setminus U}$.
$\implies \exists \epsilon > 0$ s.t. $S(x, \epsilon) \cap (X \setminus U) = \varnothing$.
$\implies S(x, \epsilon) \subseteq U$.
$\implies U$: nbd. of x. ■

- U is uniform nbd. of A
  $\iff A$ & $X \setminus U$: not in nhb. rel.

pf. ($\Longrightarrow$)

Assume U is uniform nbd. of A.
$\implies \exists \epsilon > 0$ s.t. $S(a, \epsilon) \subseteq U \ \ \forall a \in A$.
$\implies \bigcup_{a \in A} S(a, \epsilon) \subseteq U$.
$\implies \bigcup_{a \in A} S(a, \epsilon) \cap (X \setminus U) = \varnothing$. ····· (1)
$\implies \forall a \in A, \forall y \in X \setminus U$ s.t. $d(a, y) \geqslant \epsilon$.
($\because$ if not; $\exists a \in A, y \in X \setminus U$, $d(a, y) < \epsilon$, which yields contradiction to (1).)
$\implies A$ & $X \setminus U$ are not in nhb. rel. ■

---

pf. ($\Longleftarrow$)

Assume A and $X \setminus U$ is not in nhb. rel.
$\implies \exists \epsilon > 0$ s.t. $\operatorname{dist}(A, X \setminus U) = \epsilon$.
$\implies \forall a \in A$; $S(a, \epsilon) \subseteq U$.
($\because$ if not; $\exists a \in A$ s.t. $S(a, \epsilon) \not\subseteq U$,
$\iff S(a, \epsilon) \cap (X \setminus U) \neq \varnothing$,
$\implies \exists y \in X \setminus U$ s.t. $d(a, y) < \epsilon$,
contrd. to $\operatorname{dist}(A, X \setminus U) = \epsilon$.)
$\implies U$ is uniform nbh. of A. ■

- If U is (unif.) nbd. of A & $B \subseteq A$ & $U \subseteq V$, then V is (unif.) nbd. of B.

pf.

Let $b \in B$. Then, since U is a nbd. of A, $\exists \epsilon > 0$ s.t. $S(b, \epsilon) \subseteq U$.
Since $U \subseteq V$, $S(b, \epsilon) \subseteq U \subseteq V$, hence we have that V is a nbd. of B. ■

- If U & V are (unif.) nbd. of A, then is so $U \cap V$.

pf.

(i) Assume U, V: nbd. of A, and let $a \in A$.
Then $\exists \epsilon_1, \epsilon_2 > 0$ s.t.

$$S(a, \epsilon_1) \subseteq U \quad \text{and} \quad S(a, \epsilon_2) \subseteq V.$$

Let $\epsilon = \min\{\epsilon_1, \epsilon_2\}$. Then we have

$$S(a, \epsilon) \subseteq U \quad \wedge S(a\,\epsilon) \subseteq V,$$

which is equivalent to $S(a, \epsilon) \subseteq U \cap V$. □

(ii) Assume U, V: unif. nbd. of A, and let $a \in A$.
Then

$$\operatorname{dist}(A, X \setminus U) = \epsilon_1, \quad \operatorname{dist}(A, X \setminus V) = \epsilon_2$$

for some $\epsilon_1, \epsilon_2 > 0$. ····· (1)

If A & $X \setminus (U \cap V)$ are in nhb. rel.,
$\implies (X \setminus U) \cup (X \setminus V)$ & A are in nhb. rel.,
$\implies (X \setminus U)$ & A: nhb. rel., or is $(X \setminus V)$ & A,
but this contradicts to (1).
Thus A & $X \setminus (U \cap V)$ is not in nhb. rel., hence we have
$U \cap V$ is unif. nbd. of A. ■

- If U is unif. nbd. of A, then U is unif. nbd. of $\overline{A}$.

pf.

By assumption, $\exists \epsilon > 0$ s.t. $\operatorname{dist}(A, X \setminus U) = \epsilon$.
But, by Thm 1.2.28, $\operatorname{dist}(A, X \setminus U) = \operatorname{dist}(\overline{A}, X \setminus U) = \epsilon$, hence $\overline{A}$ and $X \setminus U$ is not in nhb. rel.
Thus we have that U is a unif. nbd. of $\overline{A}$. ■

· Theorems about Neighborhoods

- If $|X| < +\infty$, then all subset U of X is clopen.

pf.

Let $x \in X$, and let $y \in \overline{\{x\}}$.
Then $\mathrm{dist}(y, \{x\}) = 0$, thus $x = y$, $\overline{\{x\}} = \{x\}$.
For $n = |X|$, $\bigcup_{i=1}^{n-1} \{x_i\} = X \setminus \{x_1\}$ is closed,
hence $X \setminus \bigcup_{i=1}^{n-1} \{x_i\} = \{x_1\}$ is open.
This implies $\{x_1\}$ is clopen, and so is $U \subseteq X$. ■

- For discrete metric space (X, d_D),

$$A, B\text{: in nhb. rel.} \iff A \cap B \neq \varnothing.$$

pf.

Since A and B are in nhb. rel., $\mathrm{dist}(A, B) = 0$, thus for any $\epsilon > 0$, $\exists a \in A$ and $b \in B$ s.t. $d_D(a, b) < \epsilon$.
Since $\{d_D(a, b) : a, b \in X\} = \{0, 1\}$, and ϵ is arbitrary, we have $d_D(a, b) = 0$, which implies $a = b$.
Thus $\{a\} = \{b\} \subseteq A \cap B \neq \varnothing$. ■

- For $A, B \subseteq \mathbb{R}^n$, followings are equivalent:

(a) A, B are in nhb. rel. with respect of d_E;
(b) A, B are in nhb. rel. with respect of d_M;
(c) A, B are in nhb. rel. with respect of d_S.

pf. [(a) $\Longrightarrow$ (b) $\Longrightarrow$ (c)]

Note that $\forall x, y \in \mathbb{R}^n$,

$$d_M(x, y) \leqslant d_E(x, y) \leqslant d_S(x, y),$$

and let $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ which belong to $\mathbb{R}^n$.

. . . . . .

[i: (a) $\Longrightarrow$ (b)]
Assume A, B are in nhb. rel. in d_E.
$\Longrightarrow \forall \epsilon > 0, \exists (x, y) \in A \times B$ s.t. $d_E(x, y) < \epsilon$.
$\therefore d_M(x, y) \leqslant d_E(x, y) < \epsilon$,
thus A, B are in nhb. rel. in d_M. □

. . . . . .

[ii: (b) $\Longrightarrow$ (c)]
Assume A, B are in nhb. rel. in d_M.
$\Longrightarrow \forall \epsilon > 0, \exists (x, y) \in A \times B$ s.t. $d_M(x, y) < \epsilon/n$.

$$\Longrightarrow \quad d_S(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$
$$\leqslant n \cdot \max\{|x_i - y_i| : 1 \leqslant i \leqslant n\}$$
$$< n \cdot \epsilon/n = \epsilon.$$

Thus, A, B are in nhb. rel. in d_S. ■

pf. [(c) $\Longrightarrow$ (a)]

Assume A, B are in nhb. rel. in d_S.
$\Longrightarrow \exists \forall \epsilon > 0, \exists (x, y) \in A \times B$ s.t. $d_S(x, y) < \epsilon$.
Since $d_E(x, y) \leqslant d_S(x, y)$ $\exists \forall (x, y) \in A \times B$,
$d_E(x, y) \leqslant d_S(x, y) < \epsilon$,
thus A, B is in nhb. rel. in d_E. ■

· Subspace

- **Definition**

: For a metric space (X, d) and a subset Y of X, $(Y, d_{|_{Y \times Y}})$ is a subspace of (X, d) defined by Y.

- **Properties** ($\underline{X}$: subspace of $\underline{Y}$, $A, B \subseteq X$, $x \in X$)
  - A, B: nhb. rel. on $\underline{X} \iff A, B$: nhb. rel. on $\underline{Y}$.
  - $x \in \overline{A}$ in $\underline{X} \iff x \in \overline{A}$ in $\underline{Y}$.
  - B: (unif.) nbh. of A in $\underline{X}$
    $\iff \exists U$: (unif.) nbh. of A on $\underline{Y}$ s.t. $B = U \cap X$.
  - A: open/closed in $\underline{X}$
    $\iff \exists O$: open/closed in $\underline{Y}$ s.t. $A = O \cap X$.

§1.3. Sequences in Metric Space

· Sequences

- **Definition**

: For a map $x\colon \mathbb{N} \longrightarrow X$, $n \longmapsto x(n) =: x_n$, the range of x, i.e., $(x_n) := \{x_n : n \in \mathbb{N}\}$ is a sequence in X.

- **Convergence of sequences**
  - (x_n) converges to x, if for any U: nbd. of x, $\exists n \in \mathbb{N}$ s.t. $m \geqslant n$ implies $x_m \in U$.
  - We denoting that (x_n) converges to x by
    $$(x_n) \longrightarrow x \quad \text{or} \quad x = \lim_{n \to \infty} x_n.$$
  - We call x the limit (value) or convergent value, if $(x_n) \longrightarrow x$.

- **Accumulation point**

: $\forall U$: nbd. of x and $\forall n \in \mathbb{N}$, $\exists m \geqslant n$ s.t. $x_m \in U$.

(Limit value vs. Accumulation point
: Limit: $|\{m : \exists U \text{ s.t. } x_m \neq U\}| \in \mathbb{N}$, but
Acc. pt.: $|\{m : \exists U \text{ s.t. } x_m \neq U\}| \in \mathbb{N} \cup \{+\infty\}$.)

· Theorems

- $(x_n) \longrightarrow x$ in $\underline{X}$;
$\iff (d(x, x_n)) \longrightarrow 0$ in $\underline{\mathbb{R}}$;
$\iff \forall\forall \epsilon > 0$; $\exists n \in \mathbb{N}$ s.t. $m \geqslant n$ implies $d(x, x_m) < \epsilon$.

pf.

[(a) $\Longrightarrow$ (c)] Let $\epsilon > 0$.
Then $\exists n \in \mathbb{N}$ s.t. $m \geqslant n$ implies $x_m \in S(x, \epsilon)$.
Then we have $d(x, x_m) < \epsilon$ for all $m \geqslant n$. □

. . . . . .

[(c) $\Longrightarrow$ (a)]
Let U be a nbd. of x. Then $\exists \epsilon > 0$ s.t. $S(x, \epsilon) \subseteq U$.
For m in the assumption, $d(x, x_m) < \epsilon$, which is equivalent to $x_m \in S(x, \epsilon)$.
Hence, $m \geqslant n$ implies $x_m \in U$, thus $(x_n) \longrightarrow x$. ■

- x is an acc. pt. of (x_n) in $\underline{X}$;
$\iff 0$ is an acc. pt. of $(d(x, x_n))$ in $\underline{\mathbb{R}}$;
$\iff$ Given $\epsilon > 0$; f.e. $n \in \mathbb{N}$, $\exists m \geqslant n$ s.t. $d(x, x_m) < \epsilon$.

- If $(x_n) \longrightarrow x$,
x is a unique acc. pt. and also a unique limit value.

pf.

Let $\epsilon > 0$. Then $\exists n \in \mathbb{N}$ s.t. $d(x, x_m) < \epsilon/2$
for $m \geqslant n$. Assume that $y \in \mathbb{R}$ is an acc. pt. of (x_n).
Then $\exists k \geqslant n$ s.t. $d(y, x_k) < \epsilon/2$, thus

$$\begin{aligned} d(x, y) &\leqslant d(x, x_k) + d(x_k, y) \\ &< (\epsilon/2) \cdot 2 = \epsilon. \end{aligned}$$

hence we have $d(x, y) < \epsilon$, $\therefore\ x = y$. ■

- If $(x_n) \longrightarrow x$ and $(y_n) \longrightarrow y$,
then $(d(x_n, y_n)) \longrightarrow d(x, y)$.

pf.

Let $\epsilon > 0$. Then $\exists n \in \mathbb{N}$ s.t. $m > n$ implies

$$\min\{d(x, x_m), d(y, y_m)\} < \epsilon/2.$$

$$\begin{aligned} \Longrightarrow d(x_m, y_m) &\leqslant d(x_m, x) + d(x, y) + d(y, y_m) \\ &< d(x, y) + \epsilon, \\ d(x, y) &\leqslant d(x, x_m) + d(x_m, y_m) + d(y_m, y) \\ &< d(x_m, y_m) + \epsilon. \end{aligned}$$

Hence we have

$$|d(x_m, y_m) - d(x, y)| < \epsilon \quad \text{for } m \geqslant n,$$

thus $(d(x_n, y_n)) \longrightarrow d(x, y)$. ■

- For $a \geqslant 0$, if $d(x_n, y_n) \leqslant a$ for each n,
and $(x_n) \longrightarrow x$ and $(y_n) \longrightarrow y$, then $d(x, y) \leqslant a$.

pf.

By the assumption, $(d(x_n, y_n)) \longrightarrow d(x, y)$.
Thus for $\epsilon > 0$, $\exists n \in \mathbb{N}$ s.t. $m \geqslant n$ implies

$$d(x, x_m) < \epsilon/2, \quad d(y, y_m) < \epsilon_2.$$

$$\begin{aligned} \Longrightarrow d(x, y) &\leqslant d(x, x_m) + d(x_m, y_m) + d(y_m, y) \\ &< a + \epsilon. \end{aligned}$$

Since ϵ is arbitrary, we have $d(x, y) \leqslant a$. ■

- **Subsequential Limits**

· Let (x_{n_i}) be a subsequence of (x_n). Then,
(i) $(x_n) \longrightarrow x \implies (x_{n_i}) \longrightarrow x$.
(ii) x: acc. pt. of $(x_{n_i}) \implies$ so is of (x_n).

· For $A \subseteq X$ and $x \in X$,
$x \in \overline{A}$ iff $\exists (a_n) \in A$ s.t. $(a_n) \longrightarrow x$.

pf.

[$\Longrightarrow$] Assume $x \in \overline{A}$, then $\operatorname{dist}(x, A) = 0$.
Thus for each $n \in \mathbb{N}$, $\exists a_n \in A$ s.t. $d(x, a_n) < 1/n$.
Since $(1/n) \longrightarrow 0$, we have $(a_n) \longrightarrow x$. □

$\left(\because \text{by the denceness of } \mathbb{Q},\ \exists n \in \mathbb{N} \text{ s.t. } 0 < 1/n < \epsilon.\right.$
$\left.\text{Then, } m \geqslant n \text{ implies } d(x, a_m) < 1/m < 1/n < \epsilon.\right)$

. . . . . .

[$\Longleftarrow$] Let $\epsilon > 0$, then $\exists n \in \mathbb{N}$ s.t. $d(x, a_m) < \epsilon$
for $m \geqslant n$. Then, $\operatorname{dist}(x, A) < \epsilon$,
thus $\operatorname{dist}(x, A) = 0$, $\therefore x \in \overline{A}$. ■

· Neighbor Relationship between Sequences

- **Definition**

: $(x_n), (y_n) \in X$ are in nhb. rel. if $(d(x_n, y_n)) \longrightarrow 0$.

- **Equivalent Relation**

· **Definition**

: For a set X, $R \subseteq X \times X$ is an equivalent relation if
(i) $(x, x) \in R$; *[Reflexive Condition]*
(ii) $(x, y) \in R \implies (y, x) \in R$; *[Symmetric Cd.]*
(iii) $(x, y), (y, z) \in R \implies (x, z) \in R$. *[Transitive Cd.]*

· **Equivalentness of Nhb. Rel.**

: For $\mathscr{S} = \{(x_n) : \text{a seq. on } X\}$, let $R \subseteq \mathscr{S} \times \mathscr{S}$ s.t.

$$R = \{(x_n, y_n) : (x_n), (y_n) \text{ are in nhb. rel.}\}.$$

Then, R is an equivalent relation in $\mathscr{S}$.

pf.

(i) *Reflexive Condition.*
Let (x_n) be a seq. Then (x_n) are in nhb. rel. with itself,
thus $(x_n, x_n) \in R$. □

(ii) *Symmetric Condition.*
Let $(x_n, y_n) \in R$.
$\Longrightarrow (x_n)$ & (y_n) are in nhb. rel.
$\Longrightarrow (y_n)$ & (x_n) are in nhb. rel.
$[\because\ d(x_n, y_n) = d(y_n, x_n).]$
$\Longrightarrow (y_n, x_n) \in R$. □

(iii) *Transitive Condition.*
Let $(x_n, y_n), (y_n, z_n) \in R$.
Then $0 \leqslant d(x_n, z_n) \leqslant d(x_n, y_n) + d(y_n, z_n)$,
hence $(x_n, z_n) \in R$. ■

- **Theorems**

· For a seq. (x_n) s.t. $\lim x_n = x$,
(x_n) & (y_n) are in nhb. rel. iff $(y_n) \longrightarrow x$.

· For $A, B \subseteq X$,
A & B: nhb. rel. iff $\exists (a_n) \subseteq A$, $(b_n) \subseteq B$ s.t. (a_n) & (b_n): nhb. rel.

pf.

[$\Longrightarrow$] Since $\operatorname{dist}(A, B) = 0$, for each $n \in \mathbb{N}$, $\exists a_n \in A$ and $b_n \in B$ s.t. $d(a_n, b_n) < 1/n$.
Hence (a_n) and (b_n) are in nhb. rel. □

[$\Longleftarrow$] Since (a_n) & (b_n) are in nhb. rel,
for each $\epsilon > 0$, $\exists n \in \mathbb{N}$ s.t. $d(a_n, b_n) < \epsilon$.
Then for all $\epsilon > 0$, $\operatorname{dist}(A, B) \leqslant d(a_n, b_n) < \epsilon$,
thus $\operatorname{dist}(A, B) = 0$, and A, B are in nhb. rel. ■

· For seq.s (x_n), $(y_n) \subseteq X$,
(x_n) & (y_n) are in nhb. rel. if and only if
for any $M \subset \mathbb{N}$ s.t. $|M| = +\infty$,
$\{x_m : m \in M\}$ & $\{y_m : m \in M\}$ are in nhb. rel.

§2.1. Continuous & Uniformly Cont. Ftn.

(Let $\underline{X} = (X, d)$, $\underline{X'} = (X', d')$, $\underline{X''} = (X'', d'')$ be m.spaces.)

· Continuity

- **Definition**

: For a ftn. $f : X \longrightarrow X'$, f is continuous at x,
if $x \in \overline{A}$ implies $f(x) \in \overline{f(A)}$.

- **Judging Theorems** (at a point)

: $f : X \longrightarrow X'$ is cont. at x,
$\Longleftrightarrow \forall U$: nbd. of $f(x)$; $f^{-1}(U)$ is nbd. of x,
$\Longleftrightarrow$ for (x_n) in $\underline{X}$ s.t. $(x_n) \longrightarrow x$, $(f(x_n)) \longrightarrow f(x)$,
$\Longleftrightarrow$ given $\epsilon > 0$, $\exists \delta > 0$ s.t.
$d(x, y) < \delta$ $\exists \forall y \in X$ implies $d'(f(x), f(y)) < \epsilon$.
$(\Longleftrightarrow f(S^d(x, \delta)) \subseteq S^{d'}(f(x), \epsilon).)$

pf.

[(a) $\Longrightarrow$ (d)] Let $\epsilon > 0$, and $x \in X$.
Put $B = X' \setminus S(f(x), \epsilon)$, and $A = f^{-1}(B)$.
Then, since
$$S(f(x), \epsilon) \cap f(A) = S(f(x), \epsilon) \cap f(f^{-1}(B)) \subseteq S(f(x), \epsilon) \subseteq B = \varnothing,$$
$S(f(x), \epsilon) \cap f(A) = \varnothing$, thus $f(x) \notin \overline{f(A)}$.
$\Longrightarrow x \neq \overline{A}$, since f is cont. at x.
$\Longrightarrow \exists \delta > 0$ s.t. $S(x, \delta) \cap A = \varnothing$.
$\Longrightarrow S(x, \delta) \cap f^{-1}(X' \setminus S(f(x), \epsilon)) = \varnothing$.
$\Longrightarrow S(x, \delta) \cap (X \setminus f^{-1}(S(f(x), \epsilon))) = \varnothing$.
$\Longrightarrow S(x, \delta) \subseteq f^{-1}(S(f(x), \epsilon))$.
$\Longrightarrow f(S(x, \delta)) \subseteq f(f^{-1}(S(f(x), \epsilon))) \subseteq S(f(x), \epsilon)$.
$\Longrightarrow f(S(x, \delta)) \subseteq S(f(x), \epsilon)$. □

[(d) $\Longrightarrow$ (b)] Let U be a nbd. of $f(x)$ in $\underline{X'}$.
$\Longrightarrow \exists \epsilon > 0$ s.t. $S(f(x), \epsilon) \subseteq U$,
$\exists \delta > 0$ s.t. $f(S(x, \delta)) \subseteq S(f(x), \epsilon)$ by (d).
$\Longrightarrow f(S(x, \delta)) \subseteq U$.
$\Longrightarrow S(x, \delta) \subseteq f^{-1}(U)$.
$\Longrightarrow f^{-1}(U)$ is a nbd. of x. □

[(b) $\Longrightarrow$ (c)] Let (x_n) be a seq. in $\underline{X}$ s.t. $(x_n) \longrightarrow x$.
Let U be a nbd. of $f(x)$, then $f^{-1}(U)$ is one of x.
$\Longrightarrow \exists n \in \mathbb{N}$ s.t. $m \geqslant n$ implies $x_m \in f^{-1}(U)$.
$\Longrightarrow f(x_m) \in f(f^{-1}(U)) \subseteq U$ for all $m \geqslant n$.
$\Longrightarrow (f(x_n)) \longrightarrow f(x)$. □

[(c) $\Longrightarrow$ (a)] Let $x \in \overline{A}$.
$\Longrightarrow \exists (a_n)$ in $\underline{X}$ s.t. $(a_n) \longrightarrow x$ by Thm 1.3.10(*).
$\Longrightarrow$ By (c), $(f(a_n)) \longrightarrow f(x)$, $\therefore f(x) \in \overline{f(A)}$ by (*).
■

- **Judging Theorems** (on $\underline{X}$)

: $f\colon \underline{X} \longrightarrow \underline{X'}$ is cont. ftn.,

$\iff \forall A$: open set on $\underline{X'}$, $f^{-1}(A)$: open on $\underline{X}$.

$\iff \forall B$: closed set on $\underline{X'}$, $f^{-1}(B)$: closed on $\underline{X}$.

pf.

[(a) $\Longrightarrow$ (b)] For an open subset A of X', A is nbd. of all $f(x) \in A$ in $\underline{X'}$.

By f is cont., and Thm 2.1.2(b), $f^{-1}(A)$ is nbd. of all $x \in f^{-1}(A)$ in $\underline{X}$.

Hence we have $f^{-1}(A)$ is open on $\underline{X}$. □

. . . . . .

[(b) $\Longrightarrow$ (c)] For an closed subset A of X', $B := X' \setminus A$ is an open set on $\underline{X}$.

By (b), $f^{-1}(B)$ is an open set, thus $\overline{f^{-1}(A)} = \overline{X \setminus f^{-1}(B)} = X \setminus f^{-1}(B) = f^{-1}(A)$, so we have $f^{-1}(A)$ is an open set on $\underline{X}$. □

. . . . . .

[(c) $\Longrightarrow$ (a)] For $A \subseteq X$, $x \in X$, assume $f(x) \notin \overline{f(A)}$.

Then $x \notin f^{-1}(\overline{f(A)})$, but since $A \subseteq f^{-1}(\overline{f(A)})$ and by (c), $f^{-1}(\overline{f(A)})$ is closed.

Thus $\overline{A} \subseteq \overline{f^{-1}(\overline{f(A)})} = f^{-1}(\overline{f(A)})$, $x \notin \overline{A}$.

By contrapositioning this, we have $x \in \overline{A} \Longrightarrow f(x) \in \overline{f(A)}$, thus f is cont. ■

· Uniform Continuity

- **Definition**

: A function $f\colon \underline{X} \longrightarrow \underline{X'}$ is uniformly continuous, if A, B: nhb. rel. in $\underline{X} \Longrightarrow$ so is $f(A), f(B)$ in $\underline{X'}$.

- **Judging Theorem**

: $f\colon \underline{X} \longrightarrow \underline{X'}$ is unif. cont. ftn.

$\iff$ For U: unif. nbd. of A in $\underline{X'}$, $f^{-1}(U)$: unif. nbd. unif. nbd. of $f^{-1}(A)$ in $\underline{X}$.

$\iff$ For (x_n), (y_n): in nhb. rel. in $\underline{X}$, $(f(x_n))$, $(f(y_n))$: in nhb. rel. in $\underline{X'}$.

$\iff$ Given $\epsilon > 0$, $\exists \delta > 0$ s.t. $x, y \in X$ & $d(x, y) < \delta$ implies $d'(f(x), f(y)) < \epsilon$.

pf.

[(a) $\Longrightarrow$ (c)] For seqs. (x_n), (y_n)in $\underline{X}$, suppose $(f(x_n))$, $(f(y_n))$: not in nhb. rel. in $\underline{X'}$.

Then $\exists M \subseteq \mathbb{N}$ s.t. $|M| = +\infty$ and $\{f(x_m) : m \in M\}\&\{f(y_m) : m \in M\}$: not in nhb. rel.

By (a), $\{x_m : m \in M\}\&\{y_m : m \in M\}$: not in nhb. rel, thus (x_n) & (y_n): not in nhb. rel.

By contrapositioning this, we can show (c) stands. □

. . . . . .

[(c) $\Longrightarrow$ (d)] If ¬(d), for each $n \in \mathbb{N}$, $\exists x_n, y_n \in X$ s.t.

$$d(x_n, y_n) < 1/n \quad \& \quad d'(f(x_n), f(y_n)) \geqslant \epsilon$$

for some $\epsilon > 0$.

Then $(f(x_n))$ & $(f(y_n))$ are not in nhb. rel., while (x_n) & (y_n) are in nbh. rel., thus ¬(c) stands.

So we have ¬(d) $\Longrightarrow$ ¬(c), hence (c) $\Longrightarrow$ (d). □

. . . . . .

[(d) $\Longrightarrow$ (b)] If U: unif. nbh. of A in $\underline{X'}$, then $\exists \epsilon > 0$ s.t. $\operatorname{dist}(A, X' \setminus U) = \epsilon$. ····· (1)

By (d), for such ϵ, $\exists \delta > 0$ s.t.

$$d(x, y) < \delta \quad \text{implies} \quad d'(f(x), f(y)) < \epsilon$$

for all $x, y \in X$.

Thus, $\forall x \in f^{-1}(A)$, $\forall y \in X \setminus f^{-1}(U) = f^{-1}(X' \setminus U)$, since $d'(f(x), f(y)) \not< \epsilon$ by (1), $d(x, y) \geqslant \delta$, hence $\operatorname{dist}(f^{-1}(A), X \setminus f^{-1}(U)) \geqslant \delta > 0$, therefore we have $f^{-1}(A)$ is a unif. nbd. of $f^{-1}(U)$. □

. . . . . .

[(b) $\Longrightarrow$ (a)] Let $A, B \subseteq X$, and assume $f(A)$ & $f(B)$ be not in nbh. rel.

Then $U = X' \setminus f(B)$ is unif. nbh. of $f(A)$. By (b), $f^{-1}(U)$ is unif. nbh. of $f^{-1}(f(A))$.

Since $A \subseteq f^{-1}(f(A))$ and $f^{-1}(U) = X \setminus f^{-1}(f(B)) \subseteq X \setminus B$, $X \setminus B$ is unif. nbh. of A, and therefore, A and B are not in nbh. rel.

By contrapositioning this, we can show (a) stands. ■

· Various Continuous Function

(a) Lipschitz Continuous Function

: A Lipshitz continuous function is defined by a function $f : \underline{X} \longrightarrow \underline{X'}$ such that

$$d'(f(x),\, f(y)) \leqslant r \cdot d(x,\, y) \quad \forall x,\, y \in X, \text{ f.s. } r > 0.$$

(b) Continuity of Uniformly Continuous Function

: Uniform coninuous functions are continuous.

(c) Composite Function

: For (unif.) cont. ftn. $f : \underline{X} \longrightarrow \underline{X'}$ and $g : \underline{X'} \longrightarrow \underline{X''}$, the composite function

$$g \circ f : \underline{X} \longrightarrow \underline{X''}, \quad x \longmapsto g(f(x))$$

is also (uniformly) continuous.

(d) Imbedding Function

: For $X \subseteq Y$, the imbedding function is defined by a ftn.

$$f : \underline{X} \longrightarrow \underline{Y}, \quad x \longmapsto x.$$

An imbedding function is uniformly continuous.

(e) Restriction Function

: For a subsp. Y of X and a (unif.) cont. ftn. $f : \underline{X} \longrightarrow \underline{X'}$, the restriction function

$$f_{|Y} : \underline{Y} \longrightarrow \underline{X'}, \quad y \longmapsto f(y)$$

is also (uniformly) continuous.

(f) Induced Function

: For a (unif.) cont. ftn. $f : \underline{X} \longrightarrow \underline{X'}$ and a subspace Y of X' s.t. $f(X) \subseteq Y$, the induced function by f

$$f^{|Y} : \underline{X} \longrightarrow \underline{Y}, \quad x \longmapsto f(x)$$

is also (uniformly) continuous.

pf.: (a), (b)

(a) Let $f : \underline{X} \longrightarrow \underline{X'}$ be a Lipschitz cont. ftn., and A, B be subsets of X in nbh. rel., i.e., $\operatorname{dist}(A,\, B) = 0$. Then, for some $r > 0$,

$$\begin{aligned}\operatorname{dist}(f(A),\, f(B)) &= \inf\{d'(f(x),\, f(y)) : x \in A, y \in B\} \\ &\geqslant r \inf\{d(x,\, y) : x \in A, y \in B\} = 0,\end{aligned}$$

$\therefore$ $\operatorname{dist}(f(A),\, f(B)) = 0$, so $f(A)$ & $f(B)$ are in nbh. rel. ■

(b) Let $A \subseteq X$, and $f : \underline{X} \longrightarrow \underline{X'}$ be unif. cont. function. For $x \in \overline{A}$, $\operatorname{dist}(x,\, A) = 0$, and by def. of unif. cont., $\operatorname{dist}(f(x),\, f(A)) = 0$, hence $f(x) \in \overline{f(A)}$. ■

pf.: (c), (d), (f)

(c1. unif.) Let $f : \underline{X} \longrightarrow \underline{X'}$, $g : \underline{X'} \longrightarrow \underline{X''}$ be uni. cnt. ftns. For A, $B \subseteq X$ s.t. in nbh. rel. in $\underline{X}$, then, $f(A)$ & $f(B)$ are in nbh. rel. in $\underline{X'}$, and also $g(f(A))$ & $g(f(B))$ are in nbh. rel. in $\underline{X''}$. □

(c2. cont.) Let $f : \underline{X} \longrightarrow \underline{X'}$, $g : \underline{X'} \longrightarrow \underline{X''}$ be cont. ftns, and A be an open set in $\underline{X''}$. Since g is cont., $g^{-1}(A)$ is open in $\underline{X'}$, and because f is cont., $f^{-1}(g^{-1}(A))$ is also open in $\underline{X}$. Since $f^{-1} \circ g^{-1} = (g \circ f)^{-1}$, $(g \circ f)^{-1}(A)$ is open if A is open, thus $g \circ f$ is cont. ■

(d) For $x,\, y \in X$, $d'(f(x),\, f(y)) = d'(x,\, y) = 1 \cdot d(x,\, y)$, hence f is Lipschitz cont., also unif. cont. ■

(f: cont.) Let $f : \underline{X} \longrightarrow \underline{X'}$ be a cont ftn., $\underline{Y}$ be a subspace of $\underline{X'}$ s.t. $f(X) \subseteq Y$ and $A \subseteq Y$ be open. By Thm 1.2.40(d), $\exists$ an open (in $\underline{X'}$) set O s.t. $A = O \cap Y$. Since

$$\begin{aligned}(f^{|Y})^{-1}(A) &= (f^{|Y})(O \cap Y) \\ &= (f^{|Y})^{-1}(O) \cap (f^{|Y})^{-1}(Y) \\ &= f^{-1}(O) \cap f^{-1}(Y) = f^{-1}(O) \cap X,\end{aligned}$$

and $f^{-1}(O)$ and X are open, hence $(f^{|Y})^{-1}(A)$ is open. Hence we conclude $f^{|Y} : \underline{X} \longrightarrow \underline{Y}$ is cont. ■

(f: unif.) Similar way to (f: cont.) with following: For a subspace $\underline{Y}$ of $\underline{X}$, let A, $B \subseteq Y$. B is unif. nbh. of A on $\underline{Y}$ if and only if $\exists$ a set $U \subseteq X$ s.t. unif. nbh. of A on $\underline{X}$ and $U \cap Y = B$. ■

· Operational Function

(a) $\mathrm{add}\colon \underline{\mathbb{R}^2} \longrightarrow \underline{\mathbb{R}}$, $(x, y) \longmapsto x + y$: unif. cont.

(b) $\mathrm{mult}\colon \underline{\mathbb{R}^2} \longrightarrow \underline{\mathbb{R}}$, $(x, y) \longmapsto xy$: cont. (not unif.)

(c) $\mathrm{inv}\colon \underline{\mathbb{R} \setminus \{0\}} \longrightarrow \underline{\mathbb{R}}$, $x \longmapsto x^{-1}$: cont. (not unif.)

(d) For a subsp. $\underline{X}$ of $\underline{\mathbb{R}}$ s.t. $0 \notin \overline{X}$,
$\mathrm{inv}_{|X}\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ is unif. cont.

(e) For a bounded subsp. $\underline{X}$ of $\underline{\mathbb{R}^2}$,
$\mathrm{mult}_{|X}\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ is unif. cont.

pf.

(a) For $\epsilon > 0$, let $\delta = \epsilon/2$, and assume

$$d_E((x, y), (x', y')) = \sqrt{(x - x')^2 + (y - y')^2} < \delta.$$

Then we observe both $|x - x'|, |y - y'| < \delta$.
We calculate

$$\begin{aligned} d_E(x + y, x' + y') &= |x - x' + y - y'| \\ &\leqslant |x - x'| + |y - y'| \\ &< 2\delta = \epsilon, \end{aligned}$$

hence add is a unif. cont. ftn. ■

(b) $\forall (x, y) \in \mathbb{R}$ and $\forall \epsilon > 0$, let $\delta = \min\left\{1, \dfrac{\epsilon}{|x| + |y| + 1}\right\}$,
and assume $d_E((x, y), (x', y')) < \delta$.
Then we find $|x - x'|, |y - y'| < \delta$, and $|x'| < |x| + 1$.
We derive

$$\begin{aligned} |xy - x'y'| &\leqslant |xy - x'y| + |x'y - x'y'| \\ &= |y||x - x'| + |x'||y - y'| \\ &< |y|\delta + (|x| + 1)\delta \\ &= \delta(|x| + |y| + 1) \leqslant \epsilon, \end{aligned}$$

thus mult is a cont. ftn.

But though $((n, 0))$ and $((n, 1/n))$ are in nbh. rel.,
$(\mathrm{mult}((n, 0))) = (0)$ and $(\mathrm{mult}((n, 1/n))) = (1)$ are not,
therefore mult is not a unif. cont. ftn. ■

(c) For any $x \in \mathbb{R} \setminus \{0\}$ and $\epsilon > 0$, let $\delta = \min\left\{\dfrac{|x|}{2}, \dfrac{\epsilon x^2}{2}\right\}$,
and assume $|x - y| < \delta$. Then $|y| \geqslant |x|/2$.
(because if not, $|x| \leqslant |x - y| + y < 2(|x|/2) = |x|$; contrad.)
We conclude

$$\left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|y - x|}{|x||y|} < \frac{\delta}{|x|^2/2} = \epsilon,$$

hence inv is a cont. ftn.

But though $(1/n)$ and $(1/(2n))$ are in nbh. rel.,
$(\mathrm{inv}(1/n)) = (n)$ and $(\mathrm{inv}(1/(2n))) = (2n)$ are not,
so inv is not a unif. cont. ftn. ■

- Projection, Seq. Ftn.

· Projections of $\mathbb{R}^n$

: For $i = 1, 2, \ldots, n$, the i-th projection of $\mathbb{R}^n$

$$p_i\colon \underline{\mathbb{R}^n} \longrightarrow \underline{\mathbb{R}}, \quad (x_1, x_2, \ldots, x_n) \longmapsto x_i$$

is uniformly continuous.

· Sequential Function

: Each $f_i\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ are (unif.) cont. if and only if

$$(f_1, \ldots, f_n)\colon \underline{X} \longrightarrow \underline{\mathbb{R}^n}, \quad x \longmapsto (f_1(x), \ldots, f_n(x))$$

is (unif.) cont.

pf.

Projection. For each $i \in [1, n] \cap \mathbb{N}$,

$$\begin{aligned} &|p_i(x_1, x_2, \ldots, x_n) - p_i(y_1, y_2, \ldots, y_n)| \\ &= |x_i - y_i| \\ &\leqslant d_E((x_1, x_2, \ldots, x_n), (y_1, y_2, \ldots, y_n)), \end{aligned}$$

hence p_i is Lipschitz cont., thus it is unif. cont. ■

Seq. Ftn. ($\Longrightarrow$) For each $i \in [1, n] \cap \mathbb{N}$,
$p_i \circ (f_1, f_2, \ldots, f_n) = f_i$ is (unif.) cont. □

Seq. Ftn. ($\Longleftarrow$) Given $\epsilon > 0$, for each $i \in [1, n] \cap \mathbb{N}$,
$\exists \delta_i > 0$ s.t.

$$d(x, y) < \delta \quad \text{implies} \quad |f_i(x) - f_i(y)| < \epsilon/\sqrt{n}.$$

If we set $\delta = \min\{\delta_i : i = 1, 2, \ldots, n\}$, then $d(x, y) < \delta$ implies

$$\begin{aligned} &d_E((f_1, f_2, \ldots, f_n)(x) - (f_1, f_2, \ldots, f_n)(y)) \\ &= \sqrt{\sum_{k=1}^{n} (f_k(x) - f_k(y))^2} < \sqrt{n \cdot \left(\frac{\epsilon}{\sqrt{n}}\right)^2} = \epsilon, \end{aligned}$$

thus $(f_1, f_2, \ldots, f_n)$ is unif. cont. ■

- Operations between Functions

· For (unif.) cont. ftn. $f\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ and $g\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$,
$(f + g)\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ is also (unif.) cont.

· For cont. ftn. $f\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ and $g\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$,
$(f \cdot g)\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ is also cont.

· For cont. ftn. $f\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ s.t. $0 \notin f(X)$,
$\dfrac{1}{f}\colon \underline{X} \longrightarrow \underline{\mathbb{R}}$ is also cont.

pf.

Observe that $f + g = \mathrm{add} \circ (f, g)$,
$f \cdot g = \mathrm{mult} \circ (f, g)$ and $\dfrac{1}{f} = \mathrm{inv} \circ f$,
and each ftns are (unif.) cont. if f, g is so. ■

§2.2. Equivalent Metric & Isomorphism

· Definitions ($f\colon \underline{X} \longrightarrow \underline{X'}$: bij. ftn.)

- **Isometry / Metric Isomorphism**
  : f satisfying $d(x, y) = d'(f(x), f(y))$.

- **Uniform Isomorphism**
  : f s.t. both f & f^{-1} are unif. cont.

- **Topological Isomorphism**
  : f s.t. both f & f^{-1} are cont.

- **Isomorphic Space**
  : $\underline{X}$ and $\underline{X'}$ are following,
  if there exists, between $\underline{X}$ and $\underline{X'}$,

$$\begin{cases} \text{m.isomorphism } f & \Longrightarrow \quad \text{metrically isomorphic;} \\ \text{u.isomorphism } f & \Longrightarrow \quad \text{uniformly isomorphic;} \\ \text{t.isomorphism } f & \Longrightarrow \quad \text{topologically isomorphic.} \end{cases}$$

- **Equivalent Metric**
  : Metrics d and d' on X are
  {uniformly, topologically} equivalent, if

$$\mathrm{id}_X\colon (X, d) \longrightarrow (X, d')$$

  is {uniform, topological} isomorphism.

· Theorems

- Isometry is uniform isomorphism,
  and uniform isomorphism is topologically isomorphism.

- For any $\underline{X}$, $\mathrm{id}_X\colon \underline{X} \longrightarrow \underline{X}$ is an isometry.

- If $f\colon \underline{X} \longrightarrow \underline{X'}$ is {m/u/t} isomorphism,
  then so is $f^{-1}\colon \underline{X'} \longrightarrow \underline{X}$.

- If both $f\colon \underline{X} \longrightarrow \underline{X'}$, $g\colon \underline{X'} \longrightarrow \underline{X''}$ are {m/u/t} iso.,
  then so is $g \circ f\colon \underline{X} \longrightarrow \underline{X''}$.

pf.

(a: Isometry $\Longrightarrow$ unif. iso.)
Assume $f\colon (X, d) \longrightarrow (X', d')$ is an isometry.

Then $d'(f(x), f(y)) = d(x, y)$ for any $x, y \in X$,
and also $d(f^{-1}(x'), f^{-1}(y')) = d'(x', y')$ for any $x', y' \in X'$,

Hence both f and f^{-1} are Lipschitz cont., also unif. □

(b) $d(\mathrm{id}_X(x), \mathrm{id}_X(y)) = d(\mathrm{id}_X^{-1}(x), \mathrm{id}_X^{-1}(y)) = d(x, y)$. □

(c) Trivial since $(f^{-1})^{-1} = f$. □

(d) If f, g, f^{-1}, g^{-1} are (unif.) cont.,
then so are $g \circ f$ and $f^{-1} \circ g^{-1}$, so trivial. □

- {m/u/t} iso. satisfies the equivalent relation
  on $\{\underline{X} \mid \underline{X}$ is a metric space$\}$.

- Metrics d, d' on X are unif. isomorphic, if $\exists \epsilon,\ \delta > 0$ s.t.

$$x, y \in X \quad \text{implies} \quad \epsilon \cdot d(x, y) \leqslant d'(x, y) \leqslant \delta \cdot d(x, y).$$

pf.

Let $x, y \in X$. By the assumption, we have

$$d(x, y) \leqslant \frac{1}{\epsilon} d'(x, y) \quad \text{and} \quad d'(x, y) \leqslant \delta \cdot d(x, y).$$

Then $\mathrm{id}_X\colon (X, d) \longrightarrow (X, d')$, $x \longmapsto x$ and
$\mathrm{id}_X^{-1}\colon (X, d') \longrightarrow (X, d)$, $x \longmapsto x$ are
both bij., and Lipschitz cont., so unif. cont. ■

§2.3. The expansion of (unif.) cont. ftn. on ranges $[0,1]$, $(0,1)$, $\mathbb{R}$

· (Unif.) Cont. Functions

- For any $A \subseteq X$ s.t. $A \neq \varnothing$, a ftn.

$$f : X \longrightarrow \mathbb{R}, \quad x \longrightarrow \operatorname{dist}(x, A)$$

is uniformly continuous.

pf.

For any $x, y \in X$ and $a \in A$,
$\operatorname{dist}(x, A) \leqslant d(x, a) \leqslant d(x, y) + d(y, a)$.
Then $\operatorname{dist}(x, A) - d(x, y) \leqslant d(y, a)$,
$\operatorname{dist}(x, A) - d(x, y) \leqslant \operatorname{dist}(y, A)$,
$\operatorname{dist}(x, A) - \operatorname{dist}(y, A) = f(x) - f(y) \leqslant d(x, y)$.
Similarly, $f(y) - f(x) \leqslant d(y, x)$.
Then $|f(y) - f(x)| = d_E(f(x), f(y)) \leqslant d(x, y)$,
hence f is Lipschitx [& unif.] cont. ■

- **Expansion of Indicator Function**

: For $A, B \subseteq X$ s.t. $A, B \neq \varnothing$ and $A \cap B = \varnothing$, if

$$f : A \cup B \longrightarrow [0,1], \quad x \longmapsto \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \in B \end{cases}$$

has a (unif.) cont. expansive ftn.

$$g : X \longrightarrow [0,1], \quad x \longmapsto f(x),$$

then $\overline{A} \cap \overline{B} = \varnothing$. (i.e., A, B: not in nbh. rel.)

pf.

If not, $\exists x \in \overline{A} \cap \overline{B}$,
thus $\exists (a_n)$, (b_n) in A, B, resp. s.t. $a_n, b_n \longrightarrow x$.
But $g(a_n) = f(a_n) = 0$: $f(x) = 0$,
$g(b_n) = f(b_n) = 1$: $f(x) = 1$;
contrary to f. Thus $\overline{A} \cap \overline{B} = \varnothing$. ■

- **Urysohn Lemma**

: For $A, B \subseteq X$ s.t. $A, B \neq \varnothing$ and $\overline{A} \cap \overline{B} = \varnothing$,

$$f : X \longrightarrow [0,1], \quad x \longmapsto \frac{\operatorname{dist}(x, A)}{\operatorname{dist}(x, A) + \operatorname{dist}(x, B)}$$

is a well-defined function, and satisfies following:

· f is continuous;
· f is unif. cont. if and only if A, B: not in nbh. rel.

· Uniform Convergence

- **Definition**

: For a seq. (f_n) of (unif.) cont. ftn. $f_n : X \longrightarrow [0,1]$
and a ftn. $f : X \longrightarrow [0,1]$,
(f_n) uniformly converges to f if $\forall \epsilon > 0$, $\exists m \in \mathbb{N}$ s.t.

$$n \geqslant m \text{ and } x \in X \quad \text{imply} \quad |f_n(x) - f(x)| < \epsilon.$$

- **Preservation of (unif.) cont.**

: For a seq. (f_n) of (unif.) cont. ftns $f_n : X \longrightarrow [0,1]$,
if (f_n) unif. converges to $f : X \longrightarrow [0,1]$,
then f is also (unif.) cont.

pf. (unif.)

Let f_n be unif. cont. ftns. s.t. unif. converges to f.
Then given $\epsilon > 0$, $\exists n \in \mathbb{N}$ s.t.

$$x \in X \quad \text{implies} \quad |f_n(x) - f(x)| < \epsilon/3.$$

Since f_n is unif. cont., $\exists \delta > 0$ s.t.

$$d(x, y) < \delta \ \& \ x, y \in X \quad \text{imply} \quad |f_n(x) - f_n(y)| < \epsilon/3.$$

If $d(x, y) < \delta$, then we have

$$\begin{aligned} &|f(x) - f(y)| \\ &\leqslant |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &< 3 \cdot \epsilon/3 = \epsilon, \end{aligned}$$

hence f is unif. cont. ■

· Expansion Theorem (A: closed subspace of X)

- Any (unif.) cont. ftn. $f : A \longrightarrow [0,1]$
can be expanded to a (unif.) cont. ftn. $g : X \longrightarrow [0,1]$.

- Any (unif.) cont. ftn. $f : A \longrightarrow (0,1)$
can be expanded to a (unif.) cont. ftn. $g : X \longrightarrow (0,1)$.

- Any (unif.) cont. ftn. $f : A \longrightarrow \mathbb{R}$
can be expanded to a cont. ftn. $g : X \longrightarrow \mathbb{R}$,
but cannot be to a unif. cont. ftn.

- $(0,1)$ and $\mathbb{R}$ are not uniformly isomorphic.

§3.1. Cauchy Sq. & Compl. of Metric Sp.

· Cauchy Sequence

- **Definition**

: (x_n) is a Cauchy sequence, if given $\epsilon > 0$, $\exists n \in \mathbb{N}$ s.t.

$$m,\, m' \geqslant n \quad \text{implies} \quad d(x_m,\, x_{m'}) < \epsilon.$$

- **Boundedness of Sets**

: For $A \subseteq X$, A is bounded set if

$$\operatorname{diam} A := \begin{cases} 0 & \text{if } A = \varnothing, \\ \sup\{d(a,\, b) : a,\, b \in A\} & \text{if } A \neq \varnothing \end{cases} < \infty.$$

("diam" stands for "diameter.")

- **Judging Theorems** : Followings are equivalent:
  - (x_n) is a Cauchy sequence.
  - $\lim\limits_{n\to\infty} \operatorname{diam}\{x_m : m \geqslant n\} = 0$.
  - Any two subsequences of (x_n) are in nbh. rel.

- **Theorems**
  - For a unif. cont. ftn. $f : \underline{X} \longrightarrow \underline{Y}$, if (x_n) is a Cauchy seq. on X, then so is $(f(x_n))$.
  - If (x_n) is a Cauchy sequence, then $\{x_n : n \in \mathbb{N}\}$ is bdd.
  - For seq.s (x_n), (y_n) in nbh. rel., (x_n) is a Cauchy seq. $\iff$ (y_n) is a Cauchy seq.
  - $(x_n) \longrightarrow x \iff (x_n)$ is Cauchy seq. with acc. pt. x.

pf.

(a) Let (x_{n_i}), (x_{n_j}) be subseq.s of (x_n).
Then (x_{n_i}), (x_{n_j}) are in nbh. rel. Since f is unif. cont., $(f(x_{n_i}))$, $(f(x_{n_j}))$ are also in nbh. rel. ■

(b) By Thm 3.1.3(b), $\exists n \in \mathbb{N}$ s.t. $\operatorname{diam}\{x_m : m \geqslant n\} < 1$.
Put $r = \max\{d(a_i,\, a_n) : i = 1, 2, \ldots, n\}$.
Then $\operatorname{diam}\{x_n : n \in \mathbb{N}\} < 2(r+1)$ $(< +\infty)$. ■

(c) Given $\epsilon > 0$, $\exists n_0 \in \mathbb{N}$ s.t. $\operatorname{diam}\{x_m : m \geqslant n_0\} < \epsilon/3$, $\exists n_1 \in \mathbb{N}$ s.t. $n > n_1$ implies $d(x_n,\, y_n) < \epsilon/3$.
Let $m,\, n \geqslant \max\{n_0,\, n_1\}$. Then we have

$$\begin{aligned} d(y_n,\, y_m) &\leqslant d(y_n,\, x_n) + d(x_n,\, x_m) + d(x_m,\, y_m) \\ &< 3 \cdot \epsilon/3 = \epsilon. \end{aligned}$$

■

(d⇒) $\forall \epsilon > 0$, $\exists n \in \mathbb{N}$ s.t. $d(x_m,\, x) < \epsilon/2$ for $m \geqslant n$.
For $m,\, m' \geqslant n$, $d(x_m,\, x_{m'}) \leqslant d(x_m,\, x) + d(x,\, x_{m'}) < \epsilon$. □

(d⇐) $\forall \epsilon > 0$, $\exists n_0 \in \mathbb{N}$ s.t. $\operatorname{diam}\{x_m : m \geqslant n\} < \epsilon/2$.
Since x: acc. pt., $\exists n_1 \geqslant n_0$ s.t. $d(x_{n_1},\, x) < \epsilon/2$.
Then, for $n \geqslant n_0$, $d(x,\, x_n) \leqslant d(x,\, x_{n_1}) + d(x_{n_1},\, x_n) < \epsilon$. ■

· Complete Spaces

- **Definition**

: A metrix space $\underline{X}$ is a complete space if every Cauchy seq. on $\underline{X}$ converges in $\underline{X}$.

- **Complete Subspace**

: For a subspace $\underline{A}$ of $\underline{X}$,
$\underline{A}$ is a complete space $\iff$ A is closed in $\underline{X}$.

pf.

(⟹) Let $x \in \overline{A}$. Then $\exists (a_n) \subseteq A$ s.t. $(a_n) \longrightarrow x$.
Thus (a_n) is a Cauchy seq. in $\underline{X}$ and $\underline{A}$.
By the assumption, $(a_n) \longrightarrow x \in A$ in $\underline{A}$, $\overline{A} \subseteq A$. □

(⟸) Let (a_n) be a Cauchy seq. on $\underline{A}$.
then (a_n) is also a Cauchy seq. on $\underline{X}$,
hence it converges to $x \in X$ since $\underline{X}$ is complete.
By the assumption, $x \in \overline{A} = A$, thus $\underline{A}$ is complete. ■

- $\underline{X}$ is complete if and only if nonempty subset seq. (A_n) of X s.t. $\begin{cases} \lim\limits_{n\to\infty} \operatorname{diam} A_n = 0, \\ A_1 \supseteq A_2 \supseteq \cdots \end{cases}$ satisfies $\bigcap\limits_{n\in\mathbb{N}} \overline{A_n} \neq \varnothing$.

§3.2. Expansion of (Unif.) Cont. Ftn. on Complete Space

§3.3. Baire's Category Theorem

· **Dense Set**

- **Definition**
  : For $A \subseteq X$, A is dense in X if $\overline{A} = X$.

- **Expanstion on Unif. Cont. Ftn.**
  : For a dense subset A of X and complete space Y,
  every unif. cont. ftn. $f : A \longrightarrow Y$
  can be expanded to a unif. cont. ftn. $g : X \longrightarrow Y$.

· **Baire's Category Theorem**

- **Nowhere Denseness**
  : For $A \subseteq X$, A is nowhere dense in X if $\overset{\circ}{\overline{A}} = \varnothing$.
  (ex. $\mathbb{Z}$, $\mathbb{N}$: $\forall \epsilon > 0$ and $\forall p \in \mathbb{Z}$, $S(p, \epsilon) \not\subseteq \mathbb{Z}$.
  or every finite subset of $\mathbb{R}$)

- **1st Category Set**
  : A is a 1st category set if
  $\exists$ nowhere dense subset seq. (A_n) s.t. $A = \bigcup_{n \in \mathbb{N}} A_n$.

- **2nd Category Set**
  : A is a 2nd category set if A is not in 1st category.

- **Isolated Point**
  : $x \in X$ is an isolated point if $\{x\} = \overset{\circ}{\{x\}}$.

- For an isolated point x, every $A \ni x$ is a 2nd category set.

  pf.

  Assume A is in 1st category.
  Then $\exists$ nowhere dense subset (A_n) s.t. $A = \bigcup A_n$.
  If $x \in A$, $\exists n \in \mathbb{N}$ s.t. $x \in A_n$, i.e., $\{x\} \subseteq A_n$.
  Since $\{x\} \subseteq \overline{\{x\}}$, $\{x\} = \overset{\circ}{\{x\}} \subseteq \overset{\circ}{\overline{\{x\}}} = \varnothing$: contrary.
  Thus A is in 2nd category. ■

- **Baire's Category Theorem**
  : Every nonempty complete space is a 2nd category set.
  If $A \subseteq X$ is 1st ct., then $X \setminus A$ is dense in X.

- **Categories of Typical Sets**
  · $\mathbb{Z}$, $\mathbb{N}$, $\mathbb{Q}$ are 1st category sets.
  (Note that $\overline{\mathbb{Z}} = \mathbb{Z}$, $\overline{\mathbb{N}} = \mathbb{N}$, and $\overset{\circ}{\mathbb{Z}} = \overset{\circ}{\mathbb{N}} = \varnothing$.)
  · $\mathbb{R}$ is a 2nd category set.

§3.4. Topol. Complete Space

· **Definition**
: X is a topologically complete space
if X is topologically isomorphic to complete space Y.

· **Theorems**

- If X is topol. complete space, then X is in 2nd category.
- For a topol. complete space X,
  Both closed and open subspace of X are topol. complete.

§3.5. Completion of Metric Spaces

· **Definition**
: Y is a completion of X, if
- Y is a complete space;
- X is a dense subset of Y. (i.e., $\overline{X} = Y$.)

· **Theorems**

- If Y_1, Y_2 are completions of X,
  then $\exists$ an isometry $f : Y_1 \longrightarrow Y_2$, $x \longmapsto x$ $(x \in X)$.
- If X is metrically isomorphic to a subset of a complete set,
  then X has its completition.
- Every metric space X has its competition CX.

§4.1. Totally Bounded Metric Spaces

· **Definitions** ($\underline{X}$: metric space)

- **ϵ-dense** ($A \subseteq X$)
  : A is ϵ-dense on $\underline{X}$, if given $\epsilon > 0$,
  $$\operatorname{dist}(x, A) \leqslant \epsilon \quad \text{for all } x \in X.$$

- **ϵ-net**
  : A ϵ-net of $\underline{X}$ is a finite ϵ-dense subset of $\underline{X}$.

- **Total Boundedness**
  : $\underline{X}$ is totally bounded if
  $$X = \varnothing \quad \text{or} \quad \exists \text{ an } \epsilon\text{-net on } \underline{X} \text{ for all } \epsilon > 0.$$

  · **Judging Theorems**
  : $\underline{X}$ is totally bounded
  $\iff$ Every (x_n) on $\underline{X}$ has Cauchy subseq.
  $\iff$ Every (x_n) on $\underline{X}$ has an acc. pt. on $C\underline{X}$.

· **Theorems**

- **Total Boundedness of Range**
  : If $f : \underline{X} \longrightarrow \underline{Y}$ is surjective unif. cont. ftn.
  where $\underline{X}$ is totally bounded,
  then $\underline{Y}$ is also totally bounded.

- Every totally bounded metric space is bounded.

- For unif. cont. $f : \underline{X} \longrightarrow \underline{Y}$, if $\underline{X}$ is totally bounded,
  then $\underline{f(X)}$ is [totally] bounded in $\underline{Y}$.

- **Total Boundedness of Subspaces** ($\underline{A}$: subspace of $\underline{X}$)
  · If $\underline{X}$ is totally bounded, then so is $\underline{A}$.
  · If A is dense in $\underline{X}$ and $\underline{A}$ is totally bounded,
  then $\underline{X}$ is also totally bounded.

- **Total Boundedness of Completion**
  : $\underline{X}$ is totally bounded $\iff$ $C\underline{X}$ is totally bounded.

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§4.2. Compact Metric Space

· **Definition**
: $\underline{X}$ is a compact metric space,
if every seq. (x_n) in $\underline{X}$ has its acc. pt.

· $\underline{X}$ is compact $\iff$ $\underline{X}$ is complete and totally bounded.

· For a subset A of a compact space $\underline{X}$,
$$\underline{A} \text{ is compact} \iff \overline{A} \subseteq X.$$

· For subsets A, B of compact space $\underline{X}$,
$$A, B\text{: nbh. rel.} \iff \overline{A} \cap \overline{B} \neq \varnothing.$$

· For a compact space $\underline{X}$,
if $f : \underline{X} \longrightarrow \underline{X}$ is cont., then f is unif. cont.

pf.

For A, $B \subseteq X$ which is in nbh. rel.,
$\exists x \in \overline{A} \cap \overline{B}$ because $\underline{X}$ is complete.
Since f is cont., $f(x) \in \overline{f(A)} \cap \overline{f(B)}$,
hence $f(A)$, $f(B)$: in nbh. rel., thus f is unif. cont. ■

· For a compact space $\underline{X}$,
if $f : \underline{X} \longrightarrow \underline{Y}$ is cont. & surj., then $\underline{Y}$: compact.

· For a compace space $\underline{X}$,
any top. iso. $f : \underline{X} \longrightarrow \underline{Y}$ is unif. iso.

· For a ftn. $f : \underline{X} \longrightarrow \underline{Y}$ with compact $\underline{X}$,
f is bij. & cont;
$\iff$ f is a topological isomorphism;
$\iff$ f is a uniform isomorphism.